

ARPIT TAGADE

AI & Data Science Student

✉ arpittagade5@gmail.com ☎ +91 9421426438

📍 Nagpur, Maharashtra 440003 🔗 tagadearpit.vercel.app

🌐 linkedin.com/in/tagadearpit 🐙 github.com/tagadearpit 🇮🇳 Indian



PROFESSIONAL SUMMARY

First-year AI & Data Science B.Tech student with hands-on experience in full-stack architecture. Passionate about building high-performance, real-time distributed systems. Experienced at integrating cloud-based generative AI with physical hardware and enterprise-grade web backends. Seeking to leverage deep-stack technical proficiency to solve complex engineering challenges and drive product innovation.

EDUCATION

Bachelor of Technology: Artificial Intelligence and Data Science, 2029
Wainganga College of Engineering and Management Nagpur, Maharashtra

- Current First-Year Student
- Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Discrete Mathematics, Computer Architecture.

TECHNICAL ARSENAL

Languages

Java, C, C++, JavaScript/TypeScript, SQL

Backend & Systems

Spring Boot, Spring Security, Node.js, Express, WebSockets, REST APIs, JDBC, Oracle SQL

Core Competencies

Object-Oriented Programming (OOP), Multithreading, Real-Time Bidirectional Communication, State Management, UI/UX Engineering

Frontend Architecture

React.js, Next.js (App Router), Tailwind CSS, Framer Motion, HTML5/CSS3

Hardware & IoT

ESP32, Arduino Uno, Sensor Integration, Servo Control, OLED Interfaces, BIOS Optimization

CURRENTLY LEARNING & INTERESTS

Currently Learning

Rust (Systems programming focus), TensorFlow (Deep Learning model deployment).

Interests

Low-level kernel optimization, edge computing, and distributed system design patterns.

ENGINEERING PROJECTS & ARCHITECTURE

Neosis: Real-Time Communication Platform, *Java Spring Boot, WebSockets, React, Tailwind CSS*

- Live: <https://neosis-static-site.onrender.com> | GitHub: <https://github.com/tagadearpit/Neosis.git>
- Engineered a low-latency, real-time communication platform serving live data packets via WebSockets; achieved sub-100ms message delivery performance for concurrent user sessions.
- Engineered secure user authentication and session management pipelines utilizing Spring Security.

- Designed and built a highly responsive, modern frontend interface using React and Tailwind CSS, ensuring a seamless user experience across devices.

Intelligent Developer Portfolio & AI Assistant, *Next.js, TypeScript, Tailwind CSS, Gemini API*

- Live: <https://tagadearpit.vercel.app> | GitHub: <https://github.com/tagadearpit/my-portfolio.git>
- Architected a high-performance personal portfolio utilizing the Next.js App Router and Framer Motion for advanced, hardware-accelerated UI animations.
- Developed a bespoke AI assistant using Gemini 2.5 Flash; optimized inference latency by ~30% compared to standard API implementations, providing instant context-aware technical support to site visitors.
- Streamlined application delivery and continuous integration pipelines through automated Vercel deployments.

Monika-AI: Intelligent Web Interface, *Node.js, Vanilla JS, API Integration*

- Live: <https://monika-ai-0jpf.onrender.com> | GitHub: <https://github.com/tagadearpit/Monika-AI.git>
- Developed and deployed an interactive, server-backed web application providing users with an intelligent, responsive digital interface.
- Engineered the backend infrastructure using Node.js, ensuring rapid request handling and seamless API interactions.

Project CandyRobot: IoT & AI Integration, *ESP32, C++, Hardware Engineering, Cloud AI*

- Developed a dynamic, physical robotic interface combining ESP32 microcontrollers with servo motors and 128x64 OLED displays.
- Programmed the low-level hardware in C++ to interact seamlessly with Google's Gemini API over a network, creating an intelligent hardware system capable of real-time physical and visual feedback.

TECHNICAL INITIATIVES & SYSTEMS ENGINEERING

System Infrastructure & Performance Engineering

- Developed custom system-level optimization techniques to reconfigure BIOS memory maps and suspend non-essential background daemons, resulting in a 15% increase in usable RAM and a significant reduction in system idle power consumption on high-performance workstation environments.